

Program : Diploma in Computer Application and Business Management	
Course Code : 5251	Course Title: Visual Programming
Semester : 5	Credits: No Credit
Course Category: Program Core	
Periods per week: 4 (L:3, T:1, P:0)	Periods per semester: 60

Course Objectives:

- To provide an understanding of the programming concepts using VB.NET Programming Language.
- To understand the .NET framework for creating web applications.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basic programming		Introduction to Python	1

Course Outcomes:

On completion of the course, students will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain .NET framework and know about variables, constants, operators & expressions.	13	Understanding
CO2	Summarize control structures and arrays	10	Understanding
CO3	Explain the concept of windows programming, events and exceptions	20	Understanding
CO4	Outline ADO.NET and ASP.NET	15	Understanding
	Series Exam	2	

CO - PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1			3				
CO2		3					
CO3			3				
CO4			3				

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain .NET framework and know about variables, constants, operators & expressions.		
M1.01	Define Visual Programming	1	Understanding
M1.02	List advantages and disadvantages of visual programming	1	Understanding
M1.03	Familiarize .NET Framework and explain .NET Framework architecture	2	Understanding
M1.04	Understand CLR and .Net Class Library	2	Understanding
M1.05	Compare .NET Languages VB, C#	1	Understanding
M1.06	Explain variables and constants with examples, understand data types and declaration	2	Understanding
M1.07	Explain arithmetic, relational and logical operators	2	Understanding
M1.08	Understand operator precedence, expressions and managed code	2	Understanding
Contents: Visual Programming - .NET Framework - CLR - .NET Class Library - .NET Languages - Variables - Constants - Data Types - Operators - Expressions.			
CO2	Summarize control structures and Arrays.		
M2.01	Understand If - Then, If - Then - Else Statements	2	Understanding
M2.02	Understand Select Case Statement	1	Understanding
M2.03	Understand Do While - Loop Statement	1	Understanding
M2.04	Understand For - Next Statement	1	Understanding

M2.05	Understand exit, continue and break statements	1	Understanding
M2.06	Understand Single Dimensional Arrays	2	Understanding
M2.07	Understand Multi-Dimensional Arrays	1	Understanding
M2.08	Understand Object Arrays	1	Understanding
	Series Test-I	1	
Contents: Control Statements: If Then - If Then Else - Select Case - Looping Statements: Do While - For - Exit - Continue - Arrays - Single Dimensional - Multidimensional - Object Arrays.			
CO3	Explain the concept of windows programming, events and exceptions.		
M3.01	Describe the procedure of creating Window Forms	1	Understanding
M3.02	Explain Form Attributes	1	Understanding
M3.03	Define TextBox, ComboBox, ListBox, CheckBox and RadioButton	2	Understanding
M3.04	Define Timer and explain Menu Creation	1	Understanding
M3.05	Explain Event Driven Programming	1	Understanding
M3.06	Understand Form Events - Load, Activate, Close, Keyboard Events - KeyPress, KeyDown and Mouse Events - Click, DoubleClick, MouseMove	2	Understanding
M3.07	Explain Exception Handling; compare structured and unstructured exception handling.	1	Understanding
M3.08	Explain Crystal Reports	1	Understanding
M3.09	Define Database Fields, special fields, summary fields, lines and boxes	2	Understanding
M3.10	Define Report Header and Report Footer, Page Header and Page Footer, Group Header and Group Footer	3	Understanding
M3.11	Define Formatting Fields	1	Understanding
M3.12	Define Printer and Page Settings	1	Understanding
M3.13	Describe how to create crystal reports	2	Understanding
M3.14	List advantages of crystal report	1	Understanding

Contents:

Window Forms - Form Attributes - TextBox - Label - Combo Box - List Box - Check Box - Radio Button - Timer - Menu - Dialog Boxes - MsgBox - Input Box - Events - Form Events - Keyboard Events - Mouse Events - Exceptions - Try Catch - Crystal Reports - Database Fields - Special Fields - Summary Fields - Lines - Boxes - Report Header - Report Footer - Page Header - Page Footer - Group Header - Group Header - Formatting Fields - Printer Settings - Page Settings.

CO4	Outline ADO.NET and ASP.NET.		
M4.01	Understand ADO.NET	1	Understanding
M4.02	Define Connection Object and Command Object	2	Understanding
M4.03	Define DataAdapter, DataSet, DataTable and DataRow	3	Understanding
M4.04	List advantages of ADO.NET	1	Understanding
M4.05	Explain ASP .NET	1	Understanding
M4.06	Understand to design Web Pages using ASP .NET	3	Understanding
M4.07	List ASP .NET Controls	2	Understanding
M4.08	Explain Web Forms	2	Understanding
	Series Test-II	1	

Contents:

ADO .NET - Architecture - Connection - Command - Data Adapter - DataSet - DataTable - DataRow - ASP .NET - Forms - Controls.

Text / Reference:

T/R	Book Title/Author
R1	Introduction to Visual basic.NET - NIIT, Prentice Hall of India , 2005
R2	David S. Platt, Introducing Microsoft .NET , Microsoft Press Publishers
R3	Introduction to Microsoft ASP.NET Work Book , Microsoft Press
R4	Danny Ryan & Tommy Ryan, ASP.NET , Wiley

Online Resources:

Sl.No	Website Link
1	https://www.tutorialspoint.com/vb.net
2	https://www.thecodingguys.net/tutorials/visualbasic/vb-tutorial
3	www.visual-basic-tutorials.com
4	https://www.vbtutor.net/vbtutor.html
5	https://www.vbtutor.net/index.php/visual-basic-2017-tutorial